Electrical power and energy

antocciphysics

Name: _____ **Block:** _____ (teacher only) **Mastery Rating:** _____

LEARNING TARGET Calculate electrical transfer rate and energy use.

SUCCESS CRITERIA - I can...

1. Use P=IV with watts, amps and volts.
2. Use E=Pt with seconds for joules.
3. Distinguish power from total energy.

Core: teacher selects from 1-7. **Stretch:** 8 after checking your work.

MARZANO SELF-CHECK

① Need help starting

② Can do part; need help

③ Solve and explain with allowed supports

LANGUAGE GOAL Use “power” and “voltage” to explain your answer. Sentence starters are optional.

power: rate of energy transfer

voltage: electric potential difference; energy transferred per charge

current: rate of charge flow

01 Warm-up: bring back an earlier idea

Parallel branches draw **0.3 A** and **0.5 A**. **Find the current supplied** to their junction.

Sentence starter: I found … [unit]. This value means … in this situation.

Marzano check - circle: ① ② ③

02 Worked example: explain the prediction

Reference: P = IV; E = PΔt; P = I²R = V²/R.
Constant power for E = PΔt; last equalities apply to an ohmic resistor.

Worked reasoning: A device at **6 V** drawing **2 A** receives power VI=**12 W**. Over **5 s** it receives **60 J**. Power is an energy-transfer rate.

Your turn: Draw an input-energy arrow and **label voltage**, current, power and elapsed time with distinct units.

Sentence starter: In my work, I used … to show … . I labeled it … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

03 Read and explain the diagram or equation

Two motors use the same energy but one takes half the time. **Compare average power**.

Sentence starter: Both … . They differ in … because … .

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

04 Apply the physics

A small classroom car motor draws **2 A** at **6 V** for **5 s**.
1. **Calculate input power** and energy.
2. **Label the energy-flow diagram** with voltage, current, duration and total input, using the correct units for each.

Sentence starter: I found … [unit]. This value means … in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: ① ② ③

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Continue down each column.

Sentence starters are optional. You may plan in any language.

05 Explain your reasoning

A student says a **10 W** motor always uses more energy than a **5 W** motor. Supply a time comparison that disproves this.

Sentence starter: I notice ... . This supports ... because ... .

Marzano check - circle: 1 2 3

06 Review: connect an earlier idea

A wave travels at fixed speed. A second target is **4** times farther away than the first. **Compare one-way travel times**.
Label a wave sketch to **support the calculation or comparison**.

Word help: wave = a disturbance that can carry energy and information

Sentence starter: Both ... . They differ in ... because ... .



Time or position: choose and label

Marzano check - circle: 1 2 3

07 Exit check

A **3 W** device runs for **10 s**. **Find energy**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3

08 STRETCH: test the prediction

Two devices use the same **60 J**: one runs at **10 W**, another at **20 W**. **Find their run times**.

Sentence starter: I found ... [unit]. This value means ... in this situation.

Draw and label all energy transfers across the boundary.

Chosen system

Stored energy change: _____ J

Marzano check - circle: 1 2 3